

EE 605: Error Correction Codes

Problem Set 3

1. We briefly mentioned the generator matrix of a Generalized Reed-Solomon (GRS) code. Consider $F = GF(q)$ for $q > n$, let $\alpha_1, \alpha_2, \dots, \alpha_n$ denote distinct non-zero elements of F and let $v_1, v_2, \dots, v_n$ be non-zero elements of F . The generator matrix is given by

$$G_{GRS} = \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_n^2 \\ \dots & \dots & \dots & \dots \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \dots & \alpha_n^{k-1} \end{pmatrix} \begin{pmatrix} v_1 & 0 & 0 & \dots & 0 \\ 0 & v_2 & 0 & \dots & 0 \\ 0 & 0 & v_3 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & v_n \end{pmatrix}$$

Show that the dual of a GRS code is also a GRS code with the same set of evaluation points.

2. Consider an $(n = 4, k = 2)$ Reed-Solomon code over $F = GF(5) = \{0, 1, 2, 3, 4\}$ with evaluation points comprising of all non-zero elements of F .
 - (a) Find the generator matrix and the parity check matrix corresponding to this code.
 - (b) Find the codeword corresponding to the message $(1, 1)$.
 - (c) Suppose the received word is (2040) . Demonstrate the working of the Berlekamp-Welch algorithm and find the output codeword.
3. (*singly extended* RS codes) Consider the following parity check matrix of a $[n = q, k, d]$ code over $GF(q)$:

$$H = \begin{pmatrix} 1 & 1 & \dots & 1 & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_{q-1} & 0 \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_{q-1}^2 & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \alpha_1^{q-k-1} & \alpha_2^{q-k-1} & \dots & \alpha_{q-1}^{q-k-1} & 0 \end{pmatrix}$$

where $\alpha_1, \alpha_2, \dots, \alpha_{q-1}$ are the non-zero elements of $GF(q)$. This corresponds to adding an overall parity check to an $[n = q - 1, k, d = q - k]$ GRS code with code multipliers $v_i = \alpha_i$. Show that the code is MDS and thus the additional parity check increases the minimum distance of the code by one.

4. (*MDS code*) Consider an (n, k) MDS code $\mathcal{C}$ over alphabet Σ . For any subset of indices $S \subset \{1, 2, \dots, n\}$, let $\mathcal{C}_S$ denote the set of all codewords in $\mathcal{C}$ projected onto the indices in S . Show that for any set S of size exactly k , we have $\mathcal{C}_S = |\Sigma|^k$.

Note that this implies that for an MDS code, any set of k coordinates can be considered an information set, i.e., there is a generator matrix for the code that is systematic on the columns in those positions. We can think of the positions of an information set as carrying the information, while the remaining positions are contributing redundancy.

5. (*Syndrome Decoding*) Consider an $(n, k, 3)$ Generalized Reed Solomon code $\mathcal{C}_{GRS}$ over $\mathbb{F}_q$, $q > n$ with a parity check matrix which has distinct non-zero code locators $\alpha_1, \alpha_2, \dots, \alpha_n$ and all column multipliers equal to 1 (*such a GRS is called normalized*).
Say a codeword of $\mathcal{C}_{GRS}$ is transmitted over the erasure channel and the received word is $\mathbf{y}$ with two erasures at locations i and j . Let $(S_0, S_1)^T$ be the syndrome of this word, where for the purpose of this computation the value 0 is substituted for y_i and y_j . Show that the erased entries of the codeword can be computed from the syndrome.
6. (*Dual of RS codes*) A GRS code is said to be primitive if $q = n + 1$ and the code locators are the nonzero elements of $\mathbb{F}_q$, say $\alpha_1, \alpha_2, \dots, \alpha_n$. Let $v_1, v_2, \dots, v_n$ be the column multipliers in the parity check matrix of a primitive GRS code. Show that the parity check matrix of the dual GRS code has column multipliers α_j/v_j .
7. (*RS codes: minimum distance*) Consider an (n, k, d) RS code $\mathcal{C}$ over $\mathbb{F}_q$.
 - (a) Show that for each subset S of d coordinates, there are $q - 1$ codewords with weight d whose nonzero coordinates reside in S .
 - (b) What is the total number of codewords in $\mathcal{C}$ of weight d and $d + 1$.
8. (*RS encoding and decoding*) Consider an $(n = 5, k = 3)$ Reed-Solomon code over $F = GF(7) = \{0, 1, 2, 3, 4, 5, 6\}$ with evaluation points comprising of $\{0, 1, 2, 3, 4\}$ (with polynomial evaluations in this order in the codeword).
 - (a) Find the generator matrix $\mathcal{G}$ and the parity check matrix $\mathcal{H}$ corresponding to this code, while keeping the order of the columns in $\mathcal{G}$ corresponding to the evaluation points as $\{0, 1, 2, 3, 4\}$.
 - (b) Find the codeword corresponding to the message $(2, 3, 2)$.
 - (c) Suppose the received word is (20314) . Demonstrate the working of the Berlekamp-Welch algorithm and find the output codeword.
9. (*RS encoding and decoding*) Consider an $(n = 5, k = 3)$ *systematic* Reed-Solomon code over $F = GF(5) = \{0, 1, 2, 3, 4\}$. Say we want to encode the source message $(u_0, u_1, u_2) = (2, 3, 4)$. We construct a polynomial $P(x)$ of degree at most 2 such that $(P(0), P(1), P(2)) = (u_0, u_1, u_2)$. The overall codeword is given by $\mathbf{c}_u = (P(0), P(1), P(2), P(3), P(4))$.
 - (a) Find the encoding polynomial $P(x)$.
 - (b) Find the codeword $\mathbf{c}_u$ which is transmitted
 - (c) Suppose that the first symbol $\mathbf{c}_u(0)$ is corrupted and changed to 0 in the received word. Demonstrate the working of the Berlekamp-Welch algorithm and find the output codeword.
10. (*MDS codes*) Consider an (n, k, d) MDS code $\mathcal{C}$. Modify the code $\mathcal{C}$ in the following two ways:
 - (a) *Puncturing*: Create code $\mathcal{C}_1$ by removing the i^{th} coordinate of all codewords.

- (b) *Shortening*: Create code $\mathcal{C}_2$ by selecting all codewords which have 0 in the i^{th} coordinate and then deleting that coordinate in all the selected codewords.

Show that $\mathcal{C}_1$ and $\mathcal{C}_2$ are also MDS codes.